

RECOVERING AND COMBINING PARTIAL MULTI-CAMERA EVIDENCE

[Final Report]

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1. Background and Motivation

Over the past couple of years, news reports from both the Bay Area and Southern California have described the same kind of burglary: thieves arrive carrying a consumer Wi-Fi jammer and switch it on before breaking in. The home's security cameras keep recording, but their uploads to the cloud are silently disrupted, so when the homeowner later opens the app, the footage from the minutes that matter is simply missing—namely no clear record of what happened inside, and no usable image of the intruder's face. What makes these cases especially frustrating is that the evidence was, in a sense, captured: the cameras saw the burglary. It just never survived the trip out of the house. And looking closer, the damage is not uniform, because the closer a camera sits to the jammer, the more of its data is lost.

These incidents are what motivated this project. My idea was to try the coding theory developed in Professor Jafar's research area on exactly this situation: if every camera protects its data before uploading, then even under uneven jamming, the fragments that survive across several cameras should be recoverable and combinable—enough to reconstruct more of the break-in and more of the intruder's appearance than any single damaged camera could provide. The specific tools come from coded computation: Lagrange Coded Computing (LCC) [1] shows how data encoded as evaluations of a polynomial can be rebuilt from any sufficiently large subset of those evaluations, with privacy built in through random masking; the construction descends from classical Reed–Solomon codes [3], and its batch-computation generalization, the Cross-Subspace Alignment codes of Jia and Jafar [2], ties it to the interference-alignment ideas developed in Professor Jafar's group [4]. This project builds that idea into a working simulation and measures, step by step, how much of evidence it can preserve for an investigator.

2. Problem Definition

The system model is as follows. Several cameras observe one event and, at each time step where a camera sees the intruder, it produces an evidence record: a time stamp, an estimated location, a confidence score, a small person-crop feature vector, and visual cues such as clothing color, build, height, and whether a bag is carried. Each record is uploaded over a lossy wireless channel, and the loss rate differs per camera according to its distance from the jammer. Two assumptions bound the problem: jamming is modeled purely as loss of uploaded packets, not at the radio level, and only information that a camera actually captured can ever be recovered—the protection is applied before the loss, not after. This abstraction is deliberate rather than naive: fighting the jammer at the receiver is an active research area of its own [5] [6]. This project takes whatever the physical layer fails to deliver as an erasure and asks a complementary question: how much can coding applied before the loss still preserve.

Within this model the project addresses five questions. The first four come from the proposal: how much of a protected record can be recovered as packet loss increases; whether combining partial evidence across cameras yields a clearer account than the best single camera; how a coded system compares with an uncoded one under identical

disruption; and what transmission and computation cost the redundancy introduces. The fifth arose during the work: whether splitting the protection into two layers—so that heavy jamming returns a degraded silhouette of the observation instead of nothing—preserves additional investigative value over an ordinary all-or-nothing code.

3. Simulation Setup

The test scene is a 22-by-14-meter lot with an east-facing house, observed for a 24-second burglary sampled at 2 frames per second (48 time steps). The intruder enters through the back door, moves through the living area to the study, picks up a loot bag, and exits through the front door to the driveway; the bag therefore appears only on the exit half of the route. Five cameras cover the event—three perimeter cameras (backyard, side gate, driveway) and two indoor cameras (living room, study hallway)—and together they capture 62 records of the intruder. The jammer sits at the backyard entry point, and each camera's packet-loss rate grows with its proximity: the backyard camera, 1.5 meters away, loses 84% of its packets, while the far driveway camera loses only 11%.

One aspect of the setup deserves mention because it changed the results materially. An early version of the visibility model was purely geometric, and it let perimeter cameras report sightings through the walls of the house—roughly a quarter of all sightings were physically impossible. The scene was rebuilt so that walls block camera sightlines, the two doors become see-through only when the burglar opens them (at 4.0 s and 21.5 s, derived from the route), and a glass study window stays transparent. The correction shrank the capture baseline from 93 records to 62 and made the coverage honest: the perimeter cameras own the outdoor approach and the exit, the indoor cameras own everything between, and no camera can substitute for another through a wall. The wall-free variant is retained only as an ablation switch, which Section 5 uses to test how much the conclusions depend on the geometry.

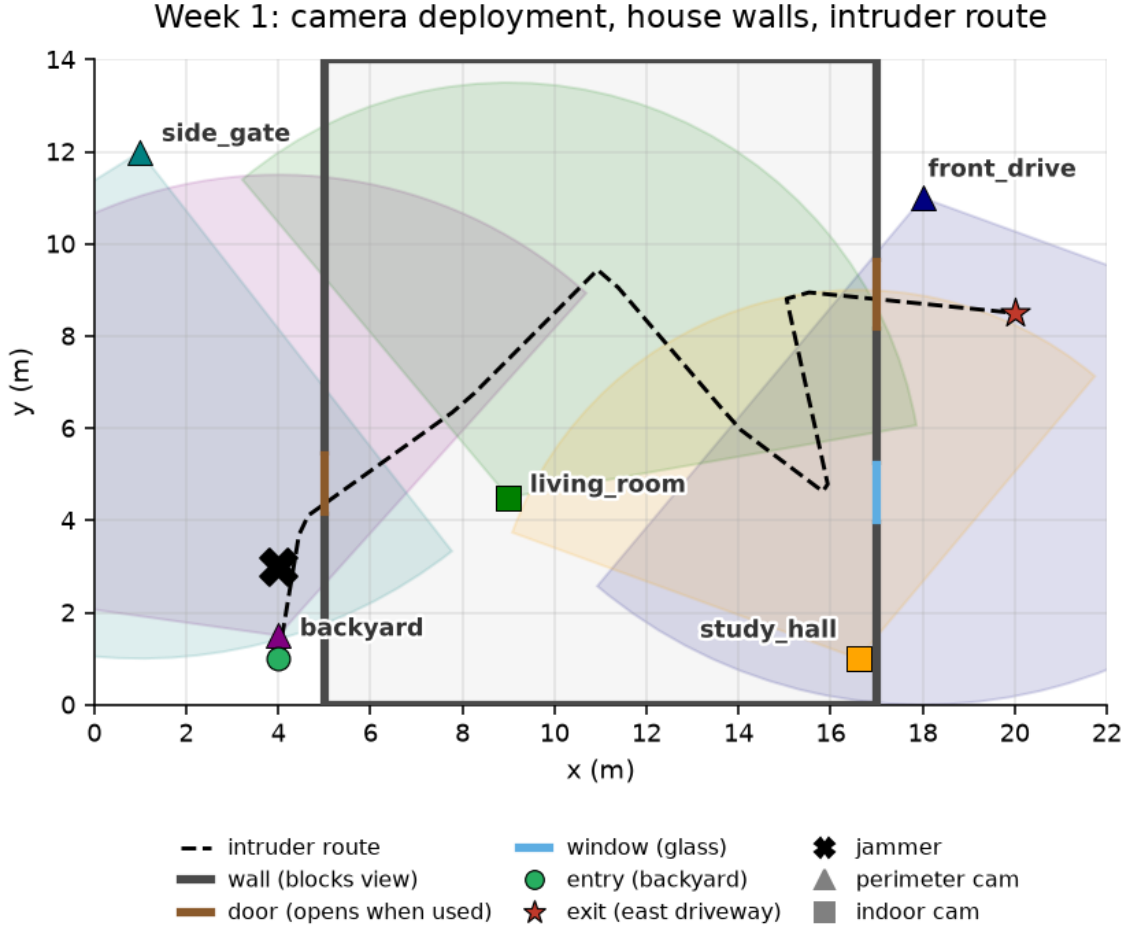


Figure 1. The simulated scene: five cameras, house walls with time-varying doors, the intruder route (dashed), and the jammer (X) at the backyard entry.

4. Method

Protection. Each evidence record is converted into 31 symbols over the finite field $GF(2003)$ and split by importance into two layers: a coarse layer of 9 symbols carrying a silhouette of the observation (rough position, height band, build, hood, a bag flag, clothing shades) and a fine layer of 22 symbols carrying the detail (precise position, exact colors, height in centimeters, the person crop). Following the LCC construction [1], each layer forms the coefficients of a polynomial, padded with $T = 2$ uniformly random masking coefficients, and both polynomials are evaluated at the same 60 points; packet i carries the pair of evaluations at point i . Decoding is polynomial interpolation, so each layer has a sharp threshold set by its degree: any 24 of the 60 packets rebuild the record in full, any 11 rebuild the silhouette, and fewer than 11 means the record is lost. The three-level outcome—full, degraded, lost—is therefore not a rule imposed on top of the code but a direct consequence of two polynomial degrees sharing one packet stream. The masks provide privacy for free, in the LCC sense: any 2 or fewer intercepted packets are statistically independent of the record.

Recovery and fusion. Every record from every camera is encoded, jammed at its own camera's loss rate, and decoded independently; the cameras cannot coordinate their encoding, because doing so would require exchanging data at capture time over exactly the link the jammer is destroying. All combination therefore happens on the server, after decoding, in three steps. Time alignment places every usable observation—full records and silhouettes alike—on the

common 48-step timeline. Association accepts an observation onto the single track only if it lies within a physically motivated gate of the previously accepted one ($3.0 \cdot \Delta t + 2.0$ meters, widened by half a meter when a position quantized to the silhouette's half-meter grid is involved); the velocity term lets the track stretch across the holes that jamming punches in the timeline while still rejecting implausible jumps. Fusion then averages positions per time step, weighting full records by their decoded confidence and silhouettes by a fixed conservative 0.25, and assembles one description of the intruder in which the highest-resolution surviving value of each attribute wins.

Experimental design. To answer the comparative questions, three settings face the same cameras, the same records, and the same channels: (1) an uncoded baseline, in which each record is sent as four plain fragments with no redundancy and counts as usable if at least three arrive—read generously, since a surviving record then enters fusion at full detail and pays no privacy overhead; (2) a coded one-layer setting, which is the identical pipeline with the silhouettes discarded, i.e., an all-or-nothing code at exactly matched overhead; and (3) the coded, layered setting. All three are evaluated by the same fusion pipeline, so the three yardsticks—evidence yield, route error against the ground-truth path, and timeline coverage—mean the same thing in every setting. Setting (2) against (1) isolates the value of coding; setting (3) against (2) isolates the value of layering. Every headline claim is averaged over ten independent runs, and one fixed reference run is used for illustration throughout. The simulation code was prepared with the assistance of Claude [7], an AI assistant by Anthropic; all design decisions were made, and all results verified, by myself.

5. Results and Analysis

Recovery thresholds. The single-record behavior matches the theory exactly: recovery probability follows the tail of a binomial distribution over surviving packets, with cliffs at 24 and 11 packets. At the heavily jammed backyard camera the expected number of surviving packets is about 9, which puts full recovery far out of reach but leaves the silhouette threshold within about half a standard deviation—precisely the regime the coarse layer was designed for. Over the whole event in the reference run, 37 of the 62 records return in full, 19 return as silhouettes, and 6 are lost; the silhouettes concentrate at the cameras nearest the jammer, where an all-or-nothing code would have returned nothing.

Fusion versus the best single camera. In the reference run the best individual camera (the living room) accounts for the intruder over 24 of 48 time steps with a route error of 1.00 m. Fusing full records from all cameras extends this to 31 steps at 1.11 m, and admitting the silhouettes extends it to 41 steps at 0.95 m—wider coverage and better accuracy at once. Across ten independent runs the layered fusion had the lower route error than the full-only fusion in ten of ten. Fusion also produces conclusions that exist in no single camera's record: the entry cameras report an empty-handed figure and the exit cameras report the same figure carrying a bag, so the theft itself becomes inferable only when the two halves of the timeline are joined.

Coded versus uncoded, and the value of layering. Table 1 summarizes the ten-run comparison of the three settings at the nominal jammer. Coding is what makes the evidence survive at all: the uncoded baseline keeps about a third of the records and covers 20 of 48 steps. Layering is what decides how the coded budget degrades: the one-layer setting converts its 60 packets into 60% of the records and nothing else, while the layered setting converts the same packets into 94% usable evidence, a route error a fifth lower, and ten more covered time steps. Sweeping the jammer strength (Figure 2) shows the same ordering everywhere: the uncoded yield decays immediately; the one-layer yield holds only until about 33% average loss and then slides down its single cliff; the layered yield stays at 100% up to about 42% average loss and is still 82% at 56%, because records falling off the fine-layer cliff land on the coarse layer instead of disappearing. Only past roughly 60% average loss do the silhouettes themselves begin to fail.

Table 1. Means over ten independent runs at the nominal jammer (ranges in parentheses). All three settings reconstructed the correct description and bag inference in every run at this jamming level.

Setting	Usable / 62	Yield	Route RMS (m)	Coverage / 48	Description & bag
(1) uncoded	22.1	36%	1.11 (0.73–1.56)	19.9	10/10 correct
(2) coded, one layer	37.2	60%	1.15 (0.93–1.35)	32.8	10/10 correct
(3) coded, layered	58.3	94%	0.93 (0.79–1.06)	42.3	10/10 correct

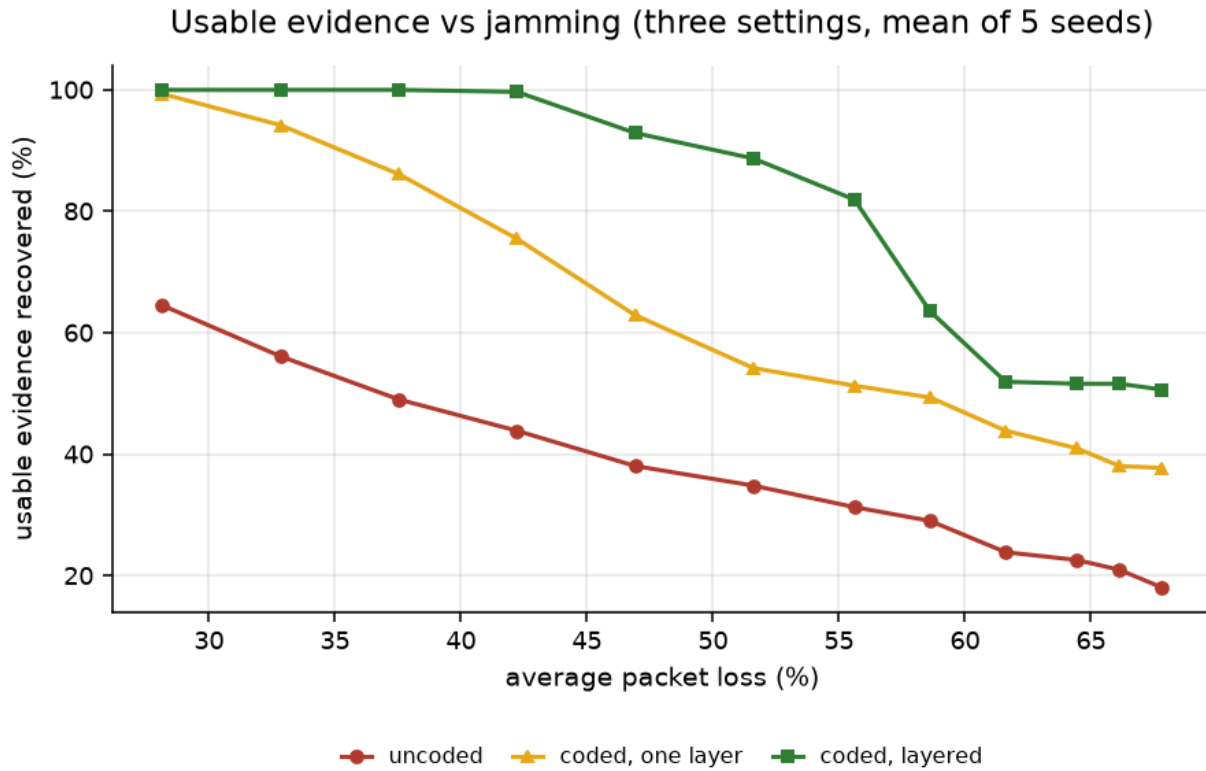


Figure 2. Usable evidence versus jammer strength for the three settings (mean of five seeds per point). The one-layer curve slides down a single cliff; the layered curve holds because records falling off the fine layer land on the coarse layer.

Under the same sweep, route quality and timeline coverage are complementary views of the same result, and neither is meaningful on its own. On accuracy (Figure 3) the layered setting keeps the lowest route error over the whole practical range, and on coverage (Figure 4) it accounts for around 40 of the 48 time steps deep into loss levels where the other two settings have already thinned out. The two figures also make the measurement caution visible: past roughly 60% average loss the uncoded curve in Figure 3 dips deceptively low—not because the uncoded route got better, but because its few surviving records cluster near the far, lightly jammed camera and cover barely a dozen steps in Figure 4.

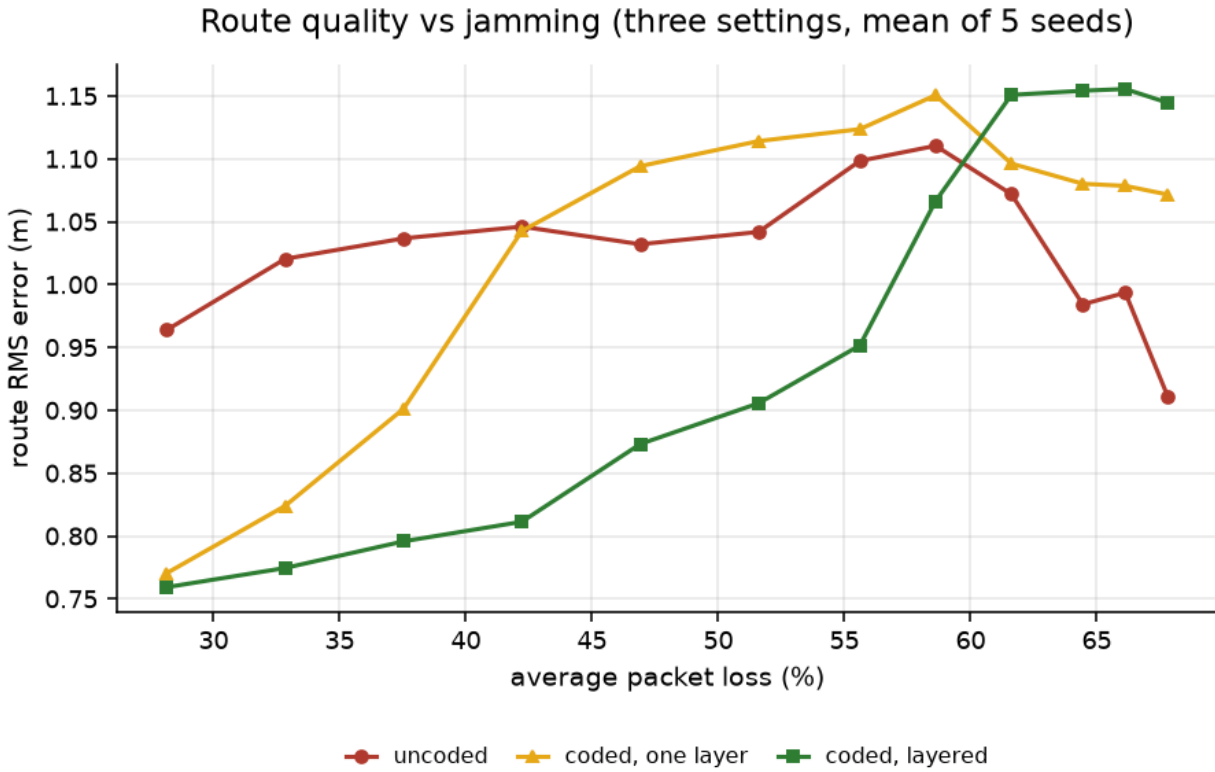


Figure 3. Route RMS error versus jammer strength. The uncoded dip at extreme loss is an artifact of shrinking coverage, not improving accuracy.

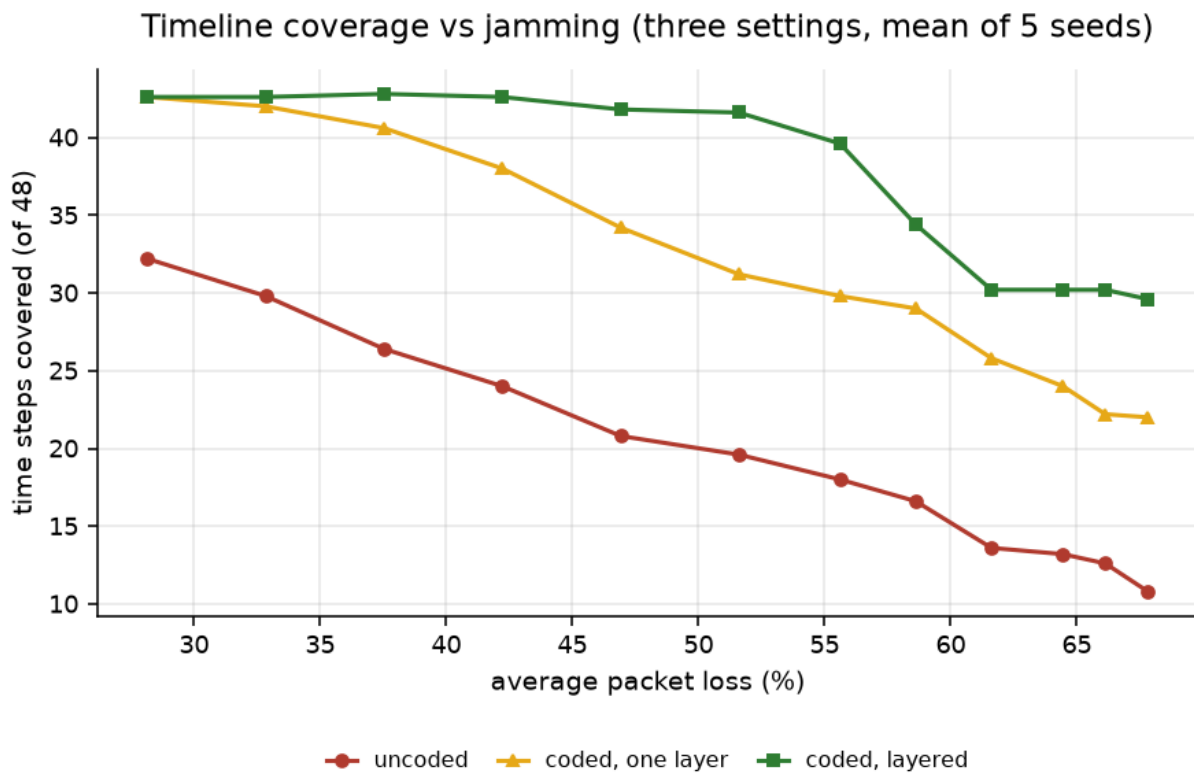


Figure 4. Timeline coverage versus jammer strength. Coverage is what disambiguates the route-error curves of Figure 3.

Cost and geometry. The resilience is not free: the layered code expands 31 symbols into 60 packets carrying 120 coded symbols, a $3.9\times$ transmission overhead, with encoding around 1 ms and decoding around 6 ms per record—negligible for evidence-sized payloads. Finally, the occlusion ablation reruns the ten-run comparison with the house walls removed (Figure 5). In the wall-free world the layered advantage in route error shrinks from +0.22 m to +0.15 m: without walls, redundant sightlines let some other camera cover for a lost record, papering over the difference between the codes. The realistic geometry is therefore not cosmetic but the honest test—the walls remove the redundancy, and that is exactly what turns the silhouettes from corroboration into the only surviving evidence over stretches of the route.

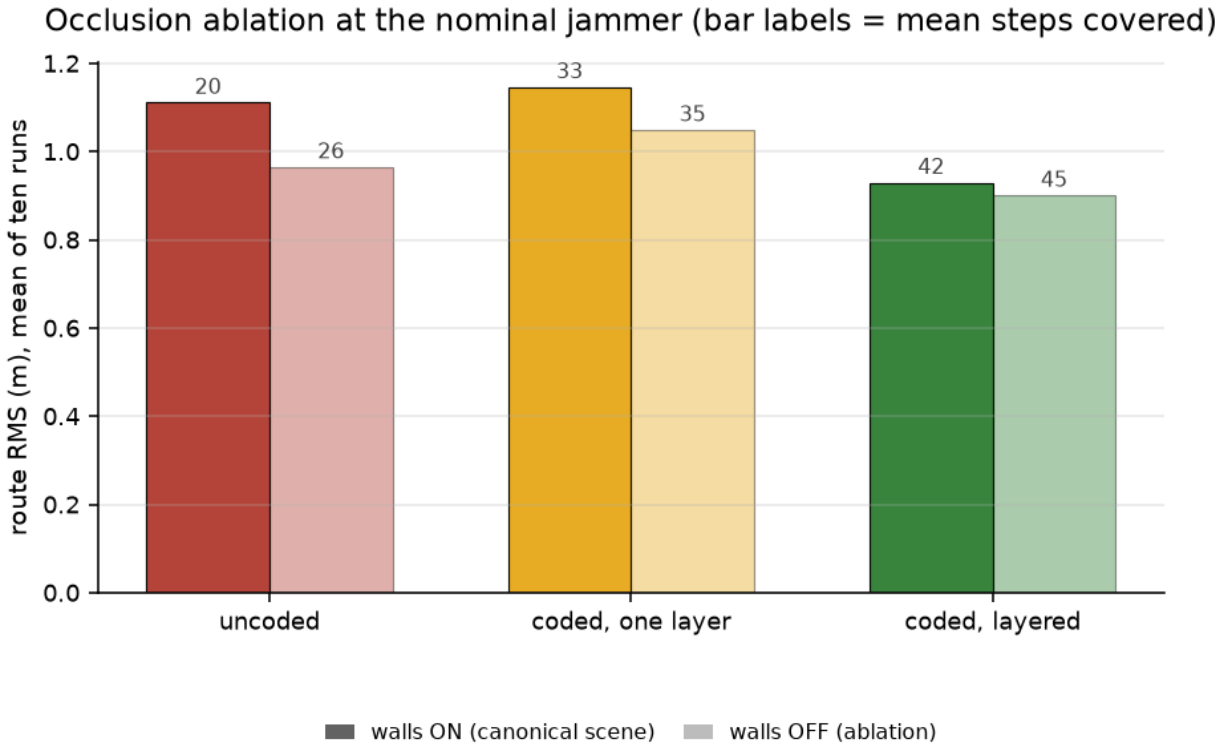


Figure 5. Occlusion ablation: mean route error per setting with the walls on (solid) and off (light); bar labels give the mean time steps covered. The layered advantage is largest in the realistic, walled scene.

6. Discussion

A few points from the project seem worth discussing beyond the headline numbers. The first is that the layering mechanism is purely algebraic: the coarse and fine layers share one packet stream and differ only in polynomial degree, so the graceful degradation costs no extra machinery. The flip side is a genuine design tension—every field added to the silhouette raises its polynomial degree and therefore its survival threshold, trading richness against survivability. Choosing what belongs in the coarse layer amounts to deciding what an investigator most needs when almost everything is destroyed; in this project that meant position, body attributes, and the bag flag, at the cost of exact colors and the image crop.

The second point is that the architecture is forced by the threat model rather than chosen for convenience. Joint encoding across cameras is circular—it would need the jammed link itself—so independence at the edge and fusion

at the server is the only consistent design. What was surprising is how little the fusion then requires: the cameras never need overlapping fields of view, only time-adjacent coverage with spatial continuity, and the most incriminating single conclusion of the run (empty-handed in, loot bag out) appears only at the fusion stage. Relatedly, recovery level and capture quality are independent axes: a record can be recovered bit-exactly and still say the shirt color was unresolved, because the camera captured it poorly—the code faithfully restores what was captured and cannot improve on it.

7. Limitations and Future Work

The clearest boundary of this project concerns the theory it draws on. Only the storage-and-recovery core of LCC is exercised here: no computation is performed in the coded domain, so the batch-computation gains of the CSA construction [2]—where the cross terms of many simultaneous multiplications are aligned into a small shared subspace, in the spirit of interference alignment [4]—have no opportunity to appear. They would become relevant in a natural extension where the server runs matching or correlation computations directly on coded evidence. The simulation itself makes simplifying choices that bound the conclusions in the same way: packet losses are independent with a fixed distance rule, there is a single intruder and a static jammer, and the person crops are synthetic rather than real video features. Burstier or adaptive jamming, real video features, and multi-target association are the corresponding directions for future work.

8. Conclusion

The project set out to show that partial multi-camera evidence can be protected, recovered, and combined after uneven jamming, and the prototype supports each of the questions posed in Section 2 with a measured answer. Recovery has sharp, predictable thresholds set by the code; fusing across cameras beats the best single camera on both accuracy and coverage; coding beats the uncoded baseline decisively at every jamming level; and the cost is a $3.9\times$ transmission overhead with millisecond-scale computation. The layered extension proved to matter most exactly where the jammer hits hardest: under heavy loss the system degrades gracefully into blurry but usable evidence instead of falling off a single cliff, delivering 94% usable evidence and the best route in every one of ten runs. The broader lesson is that a small amount of coding-theoretic structure, placed before the loss occurs, changes what an investigator can reconstruct after it.

References

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